

Peter A. Lilley

Automation Tooling Engineer (Staff Technician) · C#/.NET, C++17, BIM Tooling / CAD Drafting

Chesterfield / St. Louis, MO | pal@cadpal.net | Available on request | github.com/freeParameterized

Software engineer with a background in CAD and engineering. Builds web and mobile applications with React, Flutter, Postgres, and Firebase. Automates traditional processes with C#, C++17, and Python to simplify Civil 3D, AutoCAD, and Revit workflows. Combines AI-assisted development (Claude, ChatGPT, Gemini) with strong core engineering fundamentals from scratch.

EXPERIENCE

Automation Tooling Engineer (Staff Technician)

David Mason & Associates — St. Louis, MO

Jul 2024 - Aug 2026

- Engineered automated quality checks that cut an estimated 25% of drafting errors prior to engineer review.
- Built a generator for standard details and grading from existing survey data. Hundreds of thousands of entities; days of hand modeling became minutes.
- Wrote Dynamo automation that reads Civil 3D Excel reports and draws pipe flow levels and HGLs; about 8 hours of clicking fell to minutes.
- Cut a common multi-layout edit from several hundred clicks to a single two-second command by unifying entity search, retargeting, and batch edit.
- Built C#/.NET Civil 3D plugins, OCR ingestion of PDF plan sets, and an OpenAI pipeline with GitFlow review and CI/CD. Colleagues use the tooling daily.

Quality Engineer - Manufacturing / Precision Machining

Component Bar Products — O'Fallon, MO

Apr 2023 - Jun 2024

- Wrote inspection tooling against CMM output; validated machined geometry to GD&T specs and flagged model-to-print mismatches.
- Worked to ISO 9001 and PPAP requirements. Managed calibration and traceability of metrology assets.

Assistant Department Manager - Building Materials and Millwork

Menard, Inc. — Manchester / Ballwin, MO

Jun 2020 - Mar 2023

- Priced and estimated large commercial material orders.
- Supervised and coordinated department staff. Handled contractor-facing sales and turned requests into material lists on a deadline.

CAD Drafter

Jeff Day & Associates, LLC — Manchester, MO

2020

- Drafted architectural construction documents in AutoCAD; kept layers and the detail library to office CAD standards.

Revit/CAD Technician - Mechanical/Plumbing

Heideman & Associates, Inc. — Fenton, MO

May 2018 - Apr 2019

- Drafted mechanical and plumbing construction documents in Revit and AutoCAD, including St. Luke's, Mercy, AT&T, SLU, and American Water.
- Produced as-builts; built object libraries; standardized outdated AutoCAD details; clash detection across disciplines.

SELECTED PROJECTS

Digital Twin Pro

Flutter · Dart 3 · SQLite · Firebase · ARCore

- 3D inventory app with a hand-written renderer — Z-sorted draw queue and custom projection, no game engine.
- Voice, photo, and typed input all resolve to one typed item-to-quantity map shared by storage and export.
- Export a single tote's QR with selectable payload fields, or the whole set as a multi-page PDF for physical labels.

OraLab

TypeScript · Next.js · Java / H2 Oracle mode · Monaco

- Shipped a public Oracle SQL training lab: live engine, classic HR sample, Monaco editor, tutorials, and a docked schema graph.
- Oracle-mode SQL in-heap (DUAL, NVL/DECODE, ROWNUM, sequences, HR joins, MINUS, analytic windows). Live at freeparameterized.github.io/oracle-sandbox.

CAD integration bridge

C++17 · Dear ImGui · Windows COM

- Host process that bridges BricsCAD, AutoCAD, and Civil 3D over COM, discovers installed SDKs at runtime, and hot-loads LISP, C#, and native plugins.
- Prototype architecture that later shipped commercially.

Interactive Portfolio (this site)

React · TypeScript · Express

- Front end over a small API; builds to static files. The web page and the printable PDF render from the same JSON.

SKILLS

Languages C# / .NET, Python, C++17, Dart, TypeScript, JavaScript, SQL, Oracle SQL, LISP, VBA

Software engineering Object-Oriented Design (OOD), REST APIs, multi-threading, CI/CD pipelines, Git (GitFlow, trunk-based, pull requests), CMake, Linux, macOS, Docker, Bash

Databases and Data Systems Postgres, Oracle SQL, SQLite, Firebase, H2

CAD, geometry, and domain AutoCAD, Civil 3D, Revit, Navisworks, Dynamo, matrix math, 3D graphics, CAD plugins, typed metadata, GD&T, CMM, ISO 9001, TCP/IP, Cisco routing

EDUCATION

Ranken Technical College

A.A.S. in Building Systems Engineering Technology

2019 · Wentzville, MO

Additional coursework: Technical Spanish and Applications (Missouri S&T), Oracle Database SQL (Oracle University).